

2.1 Algorithms

How to run this unit, lesson by lesson

Contents

[Start here](#)

[Preparation and what you need](#)

[Using the three resources together](#)

[Evidence, checking and common difficulties](#)

[The lesson plans, in teaching order](#)

[Teacher answers](#)

Start here

13 lessons. A realistic estimate is **13 teaching hours**: most lessons are one hour, and a lesson with thirty or more activities takes two or three. The estimate counts transitions and assumes you teach from the front for the first fifteen minutes of each lesson.

Teach them in the order below. The order is the retrieval schedule as well as the teaching one: each starter asks about the lesson before it.

#	LESSON	HOURS	PLAN
1	Abstraction studio al-101	1	Lesson_Plans/AL_L01_LessonPlan.pdf
2	Breakdown bay al-102	1	Lesson_Plans/AL_L02_LessonPlan.pdf
3	Algorithm workshop al-103	1	Lesson_Plans/AL_L03_LessonPlan.pdf
4	Flowchart factory al-104	1	Lesson_Plans/AL_L04_LessonPlan.pdf
5	Pseudocode press al-105	1	Lesson_Plans/AL_L05_LessonPlan.pdf
6	Search corridor al-106	1	Lesson_Plans/AL_L06_LessonPlan.pdf
7	Halving hall al-107	1	Lesson_Plans/AL_L07_LessonPlan.pdf
8	Bubble bay al-108	1	Lesson_Plans/AL_L08_LessonPlan.pdf
9	Split and insert al-109	1	Lesson_Plans/AL_L09_LessonPlan.pdf
10	Fault finder al-110	1	Lesson_Plans/AL_L10_LessonPlan.pdf
11	Trace room al-111	1	Lesson_Plans/AL_L11_LessonPlan.pdf
12	End of topic test al-112-test	1	Lesson_Plans/AL_L12_TEST_LessonPlan.pdf
13	Algorithm arena al-bonus	1	Lesson_Plans/AL_BONUS_LessonPlan.pdf

Preparation and what you need

1. **Class logins, before the first lesson.** On the teacher dashboard ([teacher.html](#)), open **Class logins**, enter the class and the usernames one per line, and print the login cards. A pupil cannot sign in without one.
2. **Devices.** A laptop, a tablet or a phone will do. A headset is optional everywhere. The browser needs to be current; the experiences use WebGL.
3. **Print the worksheets** for each lesson. They are in this pack as .docx to edit and .pdf to print.
4. **Open each experience once yourself** on the teaching machine, so the images are cached before thirty pupils ask for them at the same moment.

Where progress is kept. In this build APP_CONFIG.backendUrl is empty, so logins, login cards and progress are stored on the device a pupil used, not on a server. The teacher dashboard says so itself. Keep a class on the same

devices within a unit, and treat the paper worksheet as the record that travels with the pupil.

Pupils sign in at `topics.html` with the username and PIN on their card, choose the topic and then the lesson.

Using the three resources together

Slides for the front of the room; **the experience** for reading, deciding and being marked; **the worksheet** for writing and for the record. The rule that holds them together is write first, then answer at the badge.

Do not teach the station walls and then send pupils to read the same walls. Teach the first two or three from the slides and let them read the rest; the walls are there so a pupil can re-read what you said, not so you can say it twice.

Evidence, checking and common difficulties

Collect three things from every lesson: the starter (written), the key facts and station challenges on the worksheet (written), and the score screen breakdown (copied onto the worksheet by the pupil as soon as it appears). The score screen shows the marks per station, which is what tells you where to reteach; a single total does not.

The teacher dashboard shows progress by pupil and the class's weakest stations, averaged on first attempts. Use the weakest-station list to choose what to reteach rather than asking who found it hard.

Things to plan around in this unit

- every activity is guided: nothing is practised independently — a1-101, a1-102, a1-103, a1-104
- no optional challenge activity — a1-101, a1-102, a1-103, a1-104, a1-105, a1-106, a1-107, a1-108, a1-109, a1-110, a1-111

The lesson plans, in teaching order

Each lesson has its own plan as a separate PDF in Lesson_Plans/ of this pack, so you can print the one you are teaching. What follows is the summary of each; the plan carries the timings, the hinge question, the station-by-station delivery and the adaptations.

1. Abstraction studio a1-101

What is abstraction, and why is it needed?

OUTCOMES	know what is meant by the term abstraction; know some examples of abstraction; understand why abstraction is needed in computational thinking
STATIONS AND ACTIVITIES	6 stations, 13 activities, 30 marks (13 guided, 0 independent, 0 challenge)
RESOURCES	AL_L01_AbstractionStudio_Lesson.pptx, AL_L01_AbstractionStudio_Worksheet.docx
PLAN	Lesson_Plans/AL_L01_LessonPlan.pdf

2. Breakdown bay a1-102

What is decomposition, and how does a structure diagram help?

OUTCOMES	know what is meant by problem decomposition; know the advantages of decomposition when programming; know how to produce a structure diagram
STATIONS AND ACTIVITIES	6 stations, 13 activities, 33 marks (13 guided, 0 independent, 0 challenge)
RESOURCES	AL_L02_BreakdownBay_Lesson.pptx, AL_L02_BreakdownBay_Worksheet.docx
PLAN	Lesson_Plans/AL_L02_LessonPlan.pdf

3. Algorithm workshop a1-103

What do we mean by thinking algorithmically?

OUTCOMES	understand how to solve problems by applying algorithmic thinking; be able to identify the inputs, processes and outputs of a problem
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STATIONS AND ACTIVITIES	6 stations, 13 activities, 36 marks (13 guided, 0 independent, 0 challenge)
RESOURCES	AL_L03_AlgorithmWorkshop_Lesson.pptx, AL_L03_AlgorithmWorkshop_Worksheet.docx
PLAN	Lesson_Plans/AL_L03_LessonPlan.pdf

4. Flowchart factory al-104

How can an algorithm be described without ambiguity?

OUTCOMES	know the flowchart symbols; know how to make a flowchart; understand how to construct a program from a flowchart
STATIONS AND ACTIVITIES	6 stations, 14 activities, 38 marks (14 guided, 0 independent, 0 challenge)
RESOURCES	AL_L04_FlowchartFactory_Lesson.pptx, AL_L04_FlowchartFactory_Worksheet.docx
PLAN	Lesson_Plans/AL_L04_LessonPlan.pdf

5. Pseudocode press al-105

How is an algorithm written so that anyone can follow it exactly?

OUTCOMES	know what is meant by the term pseudocode; understand how to write pseudocode; understand the OCR reference language
STATIONS AND ACTIVITIES	6 stations, 16 activities, 28 marks (15 guided, 1 independent, 0 challenge)
RESOURCES	AL_L05_PseudocodePress_Lesson.pptx, AL_L05_PseudocodePress_Worksheet.docx
PLAN	Lesson_Plans/AL_L05_LessonPlan.pdf

6. Search corridor al-106

How does a linear search work, and when is it the right choice?

OUTCOMES	understand the linear search algorithm; understand that it is inefficient but simple and needs no ordering
STATIONS AND ACTIVITIES	6 stations, 15 activities, 29 marks (11 guided, 4 independent, 0 challenge)
RESOURCES	AL_L06_SearchCorridor_Lesson.pptx, AL_L06_SearchCorridor_Worksheet.docx
PLAN	Lesson_Plans/AL_L06_LessonPlan.pdf

7. Halving hall al-107

How does a binary search work, and why is it so much faster?

OUTCOMES	understand the binary search algorithm; know the condition a list must meet for a binary search to work; understand which searching algorithm is quicker
STATIONS AND ACTIVITIES	6 stations, 14 activities, 30 marks (11 guided, 3 independent, 0 challenge)
RESOURCES	AL_L07_HalvingHall_Lesson.pptx, AL_L07_HalvingHall_Worksheet.docx
PLAN	Lesson_Plans/AL_L07_LessonPlan.pdf

8. Bubble bay al-108

How does a bubble sort work, and why is it slow?

OUTCOMES	understand the bubble sort algorithm; be able to show the steps of a bubble sort; know the advantages and disadvantages of a bubble sort
STATIONS AND ACTIVITIES	6 stations, 15 activities, 23 marks (11 guided, 4 independent, 0 challenge)

RESOURCES	AL_L08_BubbleBay_Lesson.pptx, AL_L08_BubbleBay_Worksheet.docx
PLAN	Lesson_Plans/AL_L08_LessonPlan.pdf

9. Split and insert al-109

How do merge sort and insertion sort work, and how do they compare with a bubble sort?

OUTCOMES	understand the merge sort algorithm; understand the insertion sort algorithm; be able to compare the three sorting algorithms
STATIONS AND ACTIVITIES	6 stations, 14 activities, 29 marks (11 guided, 3 independent, 0 challenge)
RESOURCES	AL_L09_SplitAndInsert_Lesson.pptx, AL_L09_SplitAndInsert_Worksheet.docx
PLAN	Lesson_Plans/AL_L09_LessonPlan.pdf

10. Fault finder al-110

What kinds of error occur when programming, and how are they found?

OUTCOMES	know what a syntax error is; know what a logic error is; be able to identify errors in code and suggest fixes
STATIONS AND ACTIVITIES	6 stations, 14 activities, 30 marks (8 guided, 6 independent, 0 challenge)
RESOURCES	AL_L10_FaultFinder_Lesson.pptx, AL_L10_FaultFinder_Worksheet.docx
PLAN	Lesson_Plans/AL_L10_LessonPlan.pdf

11. Trace room al-111

How and why do programmers use a trace table?

OUTCOMES	know what a trace table is; understand how and why trace tables help with debugging; know how to complete a trace table
STATIONS AND ACTIVITIES	6 stations, 13 activities, 55 marks (7 guided, 6 independent, 0 challenge)
RESOURCES	AL_L11_TraceRoom_Lesson.pptx, AL_L11_TraceRoom_Worksheet.docx
PLAN	Lesson_Plans/AL_L11_LessonPlan.pdf

12. End of topic test al-112-test

Your 2.1 test is done on paper. Revise with the eleven experiences and the Algorithm arena, then use this reflection sheet once it's marked.

OUTCOMES	this lesson states no outcomes of its own, which is correct for what it is: a test and reflection sheet. It assesses the outcomes taught across the unit.
STATIONS AND ACTIVITIES	0 stations, 0 activities, 0 marks (0 guided, 0 independent, 0 challenge)
RESOURCES	AL_L12_EndOfTopicTest_Reflection_Worksheet.docx
PLAN	Lesson_Plans/AL_L12_TEST_LessonPlan.pdf

13. Algorithm arena al-bonus

Two minutes of searches to run, sorts to complete and bugs to find, with speed bonuses and streaks. The challenges get harder as you go.

OUTCOMES	this lesson states no outcomes of its own, which is correct for what it is: a bonus challenge. It stretches outcomes already taught and is optional.
STATIONS AND ACTIVITIES	0 stations, 1 activities, 0 marks (0 guided, 1 independent, 0 challenge)

RESOURCES	none recorded
PLAN	Lesson_Plans/AL_BONUS_LessonPlan.pdf

Teacher answers

Model answers and marking guidance are not in this guide and not on the lesson plans. A plan is printed and left on a desk; an answer sheet is not. Where verified answers exist for this unit they are in the Teacher_Answers/ folder of this pack. Where they do not, the pack says so rather than shipping an empty file.

For the programming lessons there is no single model answer to print: a program is marked by running it against its test cases, and many different correct programs pass. The tests themselves are the marking guidance, and they are in the teacher answers.

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